ATOMIC-LEVEL MATERIALS SIMULATION

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Smearing and Tetrahedron Methods

I. The Numerical Singularity Challenge of the Dirac Delta Function

II. Mathematical Principles and Parameter Trade-offs of Smearing Methods

III. Geometric Tiling and Linear Interpolation of the Tetrahedron Method

IV. Comparative Paradigms and Algorithmic Selection Guidelines

I. The Numerical Singularity Challenge of the Dirac Delta Function

The exact mathematical expression for the Density of States relies on the ideal Dirac delta function:

$$g(E) = \frac{1}{V}\sum_{n,k} \delta[E - E_n(k)]$$

The Computational Conflict:

- **Mathematical Delta: Infinite height, zero width, total area equals one**
- **Computer Variable: Floating-point arrays require discrete, finite numerical values**

Sampling a continuous band structure with a finite *k* mesh yields discrete eigenvalues, not a continuous curve

The root of the problem lies in the definition of the Dirac delta function. In an analytical continuum, the delta function has zero width and infinite height. On a computer, however, all calculations are performed using finite arrays and discrete *k* point meshes. Diagonalizing a tight-binding or DFT Hamiltonian yields a finite set of energy eigenvalues. Without a numerical translation layer, the raw output is simply a collection of infinitely sharp spikes, which cannot be directly mapped to a smooth macroscopic property

I. The Numerical Singularity Challenge of the Dirac Delta Function

The Mechanism: Divide the energy axis into uniform intervals (bins) of width ΔE and count the eigenvalues falling inside each bin

The Inherent Structural Flaws:

- **High sensitivity to the starting point and width (ΔE) of the bins**
- **Discontinuous jumps at bin boundaries create artificial noise**
- **Highly jagged profiles require massive, unfeasible k point meshes to smooth out**

Conclusion: The simple binning approach is insufficient for high-quality scientific plots or precise thermodynamic integrations

The most intuitive solution is the binning or histogram method, where the energy range is cut into equal intervals. However, this approach introduces severe numerical artifacts. Small changes in the bin width or the grid origin can drastically alter the shape of the curve

Furthermore, it treats the continuous dispersion relation as a step-like sequence, generating high-frequency sawtooth noise. To eliminate this noise using raw histograms alone, the grid density would need to approach infinity, which is computationally prohibitive

I. The Numerical Singularity Challenge of the Dirac Delta Function

The Analytic Smearing Framework:

The Core Concept: Replace the singular, discontinuous Dirac delta function with a smooth, continuous analytical distribution function of finite width σ

The Approximation: $\delta(x) \approx \delta_{smeared}(x, \sigma)$

As the parameter σ approaches zero, the smeared function mathematically converges back to the exact Dirac delta function

The total area under the selected smoothing function must remain normalized to unity across all space

To move past the limitations of simple histograms, simulation codes implement analytic smearing.

The foundational idea is to replace the infinitely sharp delta function with a localized, well-behaved continuous function. This function spreads the weight of each discrete eigenvalue across a narrow energy window defined by the parameter σ. This transformation converts a collection of separate data points into a continuous, differentiable curve, provided the normalization remains strictly equal to one

I. The Numerical Singularity Challenge of the Dirac Delta Function

Common Smearing Formulations

Gaussian Broadening Function: $\delta_{Gaussian}(x,\sigma) = \frac{1}{\sigma\sqrt{\pi}} e^{\left(-\frac{x^2}{\sigma^2}\right)}$

Retains excellent localization properties due to rapid exponential decay

Lorentzian Broadening Function: $\delta_{Lorentz}(x,\Gamma) = \frac{1}{\pi}\frac{\Gamma}{x^2+\Gamma^2}$

Possesses broad tails; useful for modeling lifetime broadening or experimental photoemission spectra

Two mathematical shapes dominate the smearing landscape. The Gaussian function is the preferred choice for standard electronic structure calculations. Its fast exponential decay ensures that an eigenvalue only influences states in its immediate energy neighborhood. The Lorentzian function, on the other hand, features much broader tails that decay slowly. While less optimal for clean ground-state integration, the Lorentzian profile mimics real physical lifetimes and scattering processes, making it highly effective for matching experimental spectroscopy data

II. Mathematical Principles and Parameter Trade-offs of Smearing Methods

The choice of the parameter σ is a critical balance between resolution and stability

Case A: Over-smoothing (σ is too large)

Physical features, including Van Hove singularities and shallow band gaps, are artificially flattened out

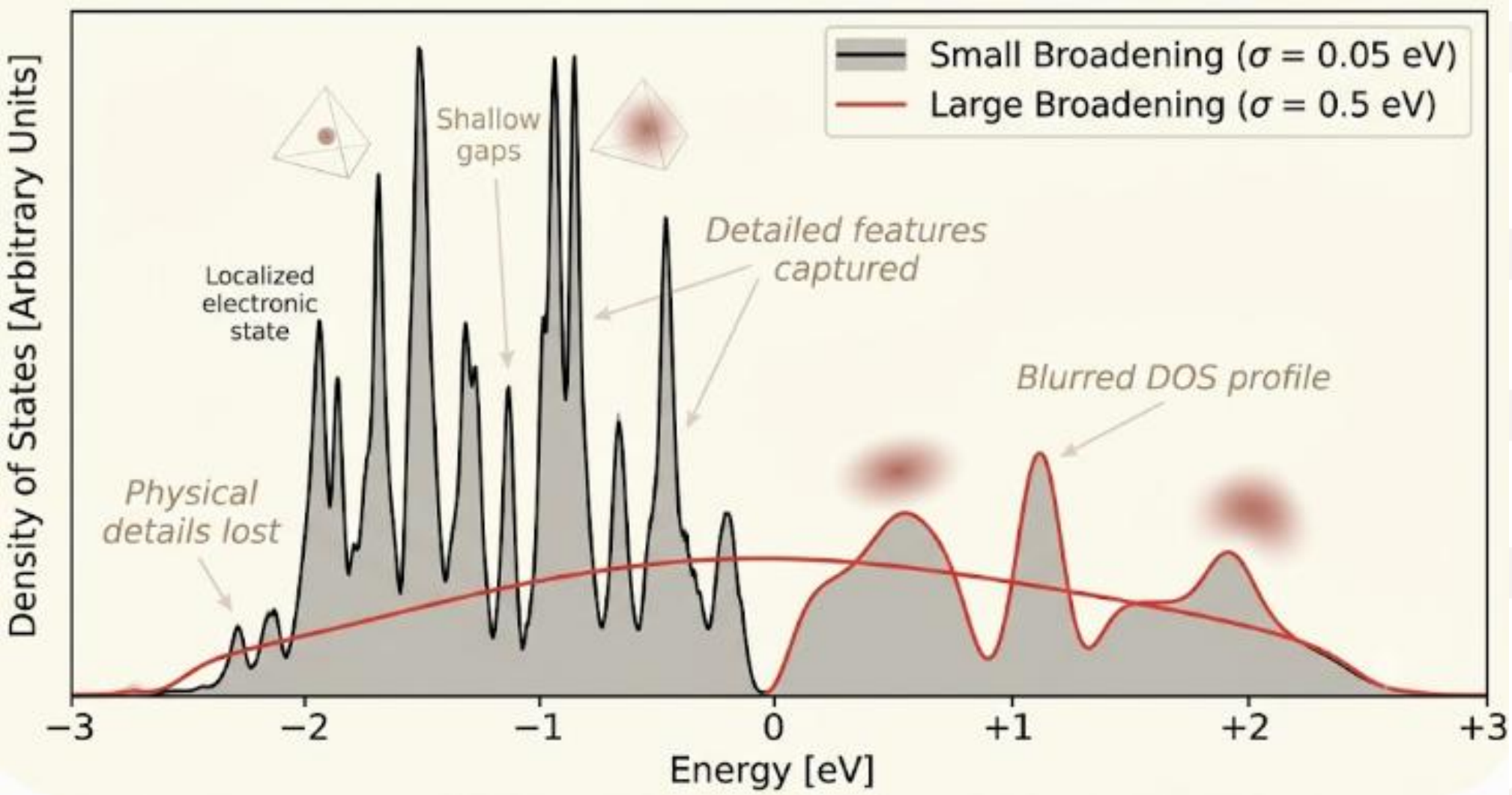
Case B: Under-smoothing (σ is too small)

The curve splits into independent, jagged peaks, exposing the underlying discrete nature of the *k* point mesh

Selecting the smearing width σ involves a direct trade-off. If σ is chosen too large, the algorithm over-smooths the data. Sharp peaks, such as Van Hove singularities, are flattened, and small semiconductor band gaps can be artificially filled with unphysical states. Conversely, if σ is too small, under-smoothing occurs.

The Gaussian tails become too narrow to bridge the gaps between adjacent discrete *k* points, causing the curve to break apart into unphysical oscillations that simply mirror the finite nature of the grid

II. Mathematical Principles and Parameter Trade-offs of Smearing Methods



II. Mathematical Principles and Parameter Trade-offs of Smearing Methods

Co-dependency of Broadening and Grid Resolution

The broadening width σ cannot be determined in isolation from the mesh spacing

The Resolution Invariant Rule: $\sigma \geq \alpha \cdot \left|\frac{\partial E(k)}{\partial k}\right| \cdot \Delta k$

Decreasing σ to capture finer physical details requires a proportional increase in the k point sampling density to prevent numerical breakdown

The parameters σ and the k mesh density are fundamentally linked. The smearing width must always be large enough to bridge the energy spacing between neighboring k points in the grid. If a research project requires high energy resolution—meaning a very small σ —the k point mesh must be made denser to compensate. Attempting to reduce the smearing width without increasing the grid density will break the continuity of the calculation and introduce severe numerical artifacts

II. Mathematical Principles and Parameter Trade-offs of Smearing Methods

The Challenge of Metallic System Integrations

In metallic systems, the electronic occupation drops discontinuously from 1 to 0 at the Fermi energy (E_F)

Standard Gaussian smearing introduces unphysical electronic states across this boundary

The Consequence: This artificial smearing perturbs the total free energy and compromises the precise evaluation of the true Fermi surface location

While simple Gaussian smearing works well for insulators and semiconductors, it encounters a serious physical bottleneck in metals. At zero temperature, a metal features a sharp, discontinuous drop in electronic occupation at the Fermi surface. Standard Gaussian smearing blurs this boundary symmetrically, which non-physically shifts the electronic density

This artificial spreading introduces an unphysical entropy contribution, corrupting total energy calculations and making it difficult to locate the exact position of the Fermi level

II. Mathematical Principles and Parameter Trade-offs of Smearing Methods

Advanced Smearing: The Methfessel-Paxton Realization

Expand the Dirac delta function using a series of Hermite polynomials:

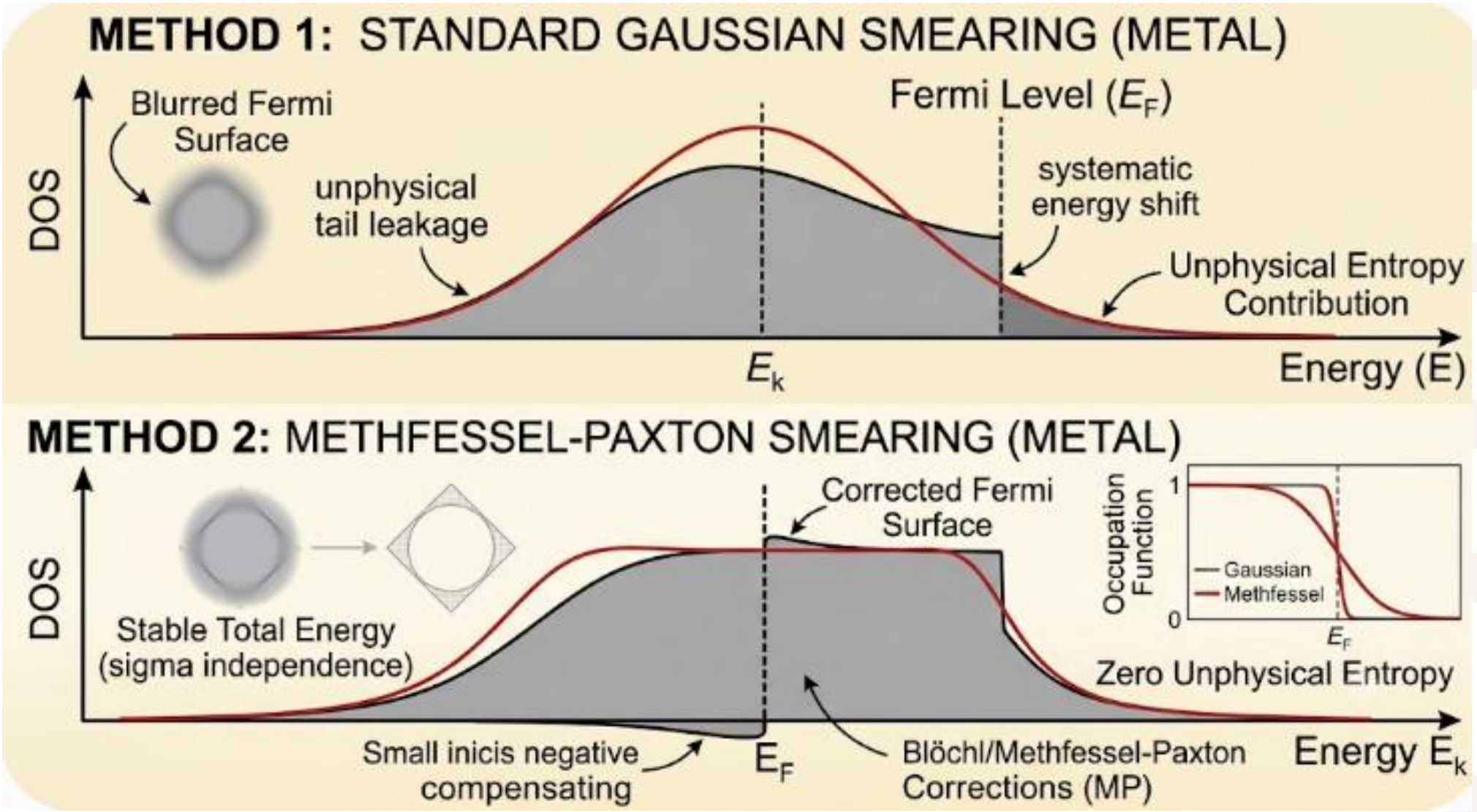
$$\delta_N(x,\sigma) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{x^2}{\sigma^2}} \sum_{m=0}^{A} A_m H_{2m}\left(\frac{x}{\sigma}\right)$$

The Critical Advantage:

- **Drastically reduces the dependence of the total energy on the smearing width**
- **Introduces compensating negative values that cancel out the first-order errors near the Fermi surface**

To resolve the metallic entropy problem, advanced techniques like the Methfessel-Paxton method are implemented. Instead of a single Gaussian profile, this approach expands the delta function using a series of even-order Hermite polynomials. This mathematical expansion introduces localized negative corrections that counteract the unphysical smoothing across the Fermi surface. As a result, the total energy calculation becomes remarkably stable against variations in σ, allowing researchers to achieve accurate metallic integrations even with moderate k point meshes

II. Mathematical Principles and Parameter Trade-offs of Smearing Methods



III. Geometric Tiling and Linear Interpolation of the Tetrahedron Method

Method 2: The Geometric Tetrahedron Method

The Core Philosophy: Eliminate arbitrary, non-physical smearing parameters entirely

The Geometric Strategy: Divide the continuous reciprocal space volume into an array of microscopic geometric tiles

- **In 3D Space: Tiled into Tetrahedrons**
- **In 2D Space: Tiled into Triangles**

Within each independent cell, the real energy bands are mapped using local linear interpolation

The second major paradigm is the tetrahedron method, which takes a geometric rather than an analytical approach. Instead of smoothing out discrete points after the calculation, this method changes how the reciprocal space is parsed from the start. The continuous Brillouin Zone is subdivided into small geometric blocks—tetrahedrons for 3*D* crystals or triangles for 2*D* structures. By treating the space as a continuous patchwork of geometric tiles, the algorithm can track states analytically without relying on arbitrary, user-defined smearing parameters

III. Geometric Tiling and Linear Interpolation of the Tetrahedron Method

Geometric Tiling Profiles in First Brillouin Zone

Every set of four neighboring points on a standard 3D Monkhorst-Pack grid forms the corners of a micro-tetrahedron volume

The exact, complex energy band structure $E(k)$ within this small volume is approximated as a flat, linear plane

The gradient of the energy remains strictly constant inside the boundaries of each individual tetrahedron:

$$\nabla_k E = Constant\ Vector$$

Let us examine the geometry of this approach. When the grid is established, every group of four adjacent points defines a unique micro-volume

The algorithm assumes that this volume is small enough that the true, curved energy band can be accurately approximated as a flat plane. Because the energy varies linearly across the cell, its gradient becomes a simple, constant vector inside that specific tetrahedron. This linear approximation enables precise, closed-form analytic integration

III. Geometric Tiling and Linear Interpolation of the Tetrahedron Method

Resolving the Delta Function via Iso-Energy Intersections

To find the DOS at a target energy *E*, the algorithm calculates how that exact constant-energy plane intersects the micro-tetrahedron volume

Depending on the target energy value, the intersection forms either a single point, a triangle, or a quadrangle

The exact area of this intersecting geometric plane can be calculated analytically using the corner energies ($E_1 \leq E_2 \leq E_3 \leq E_4$)

This process replaces the numerical delta function with exact geometric area ratios

This slide demonstrates the geometric precision of the technique. To find the density of states at an energy *E*, the code calculates how a flat plane at that exact energy intersects each individual tetrahedron. Depending on where the target energy falls relative to the corner values, the intersection cuts a precise geometric shape—a point, a triangle, or a four-sided polygon. The area of this intersecting plane can be solved exactly using analytical equations based on the corner energies. By summing these areas across all tetrahedrons, the delta function is integrated exactly without any artificial smoothing

III. Geometric Tiling and Linear Interpolation of the Tetrahedron Method

Piecewise Analytic Formulation for a Single Cell

Let the sorted corner energies of a tetrahedron be ($E_1 \leq E_2 \leq E_3 \leq E_4$)

For the Energy Range $E_1 \leq E \leq E_2$: $g_{cell}(E) = \frac{3V_{cell}}{V_{BZ}} \frac{(E-E_1)^2}{(E_2-E_1)(E_3-E_1)(E_4-E_1)}$

Similar distinct, closed-form quadratic expressions govern the subsequent energy windows ($E_2 \leq E \leq E_3$) and ($E_3 \leq E \leq E_4$)

This slide highlights the exact mathematical formulas used in the calculation. The density of states within a single cell is governed by a piecewise quadratic function of energy. For example, when the target energy lies between the lowest two corners, the density of states grows quadratically as $(E - E_1)^2$, scaled by the differences between the corner energies.

Because these equations are exact analytical solutions for a linear plane, they eliminate statistical noise and ensure that the step-like changes in slope are preserved perfectly

III. Geometric Tiling and Linear Interpolation of the Tetrahedron Method

Capturing Van Hove Singularities with Geometric Precision

Smearing Method: Smooths out sharp features, rounding off the distinct peaks and troughs of Van Hove singularities

Tetrahedron Method: Preserves the exact mathematical discontinuities in the slope ($^{dg}/_{dE}$)

It accurately reproduces critical geometric features, including local minima (M_0), saddle points (M_1, M_2), and local maxima (M_3)

The primary advantage of the tetrahedron method is its ability to capture Van Hove singularities with extreme precision. Because it avoids artificial smoothing, the sharp kinks, steps, and slope changes are preserved exactly as dictated by the crystal topology

In contrast, standard smearing techniques smooth out these critical features, rounding off the sharp peaks into broad curves. For research tasks that depend heavily on fine details at the band edges, the tetrahedron method offers a significant advantage

IV. Comparative Paradigms and Algorithmic Selection Guidelines

The Fermi Surface Intersection Bottleneck

The Complication: In a metallic system, the Fermi level cuts directly through individual tetrahedrons

A simple linear interpolation assumes the electronic occupation varies smoothly, which fails to capture the sharp discontinuity at the Fermi surface

This limitation can cause significant errors when computing properties like the Fermi velocity or metallic specific heat unless specialized corrections are applied

Despite its geometric accuracy, the basic tetrahedron method encounters difficulties when dealing with metallic systems.

When the Fermi surface cuts through a tetrahedron, the electronic occupation drops instantly from one to zero. A simple linear approximation across the cell cannot capture this sharp step change accurately. Without specialized modifications, this limitation can lead to errors when calculating properties that depend heavily on the exact state of the Fermi surface

IV. Comparative Paradigms and Algorithmic Selection Guidelines

Resolving Metallic Inaccuracies: Blöchl's Corrections

Developed by Peter Blöchl in 1994 to correct integration errors in metallic systems

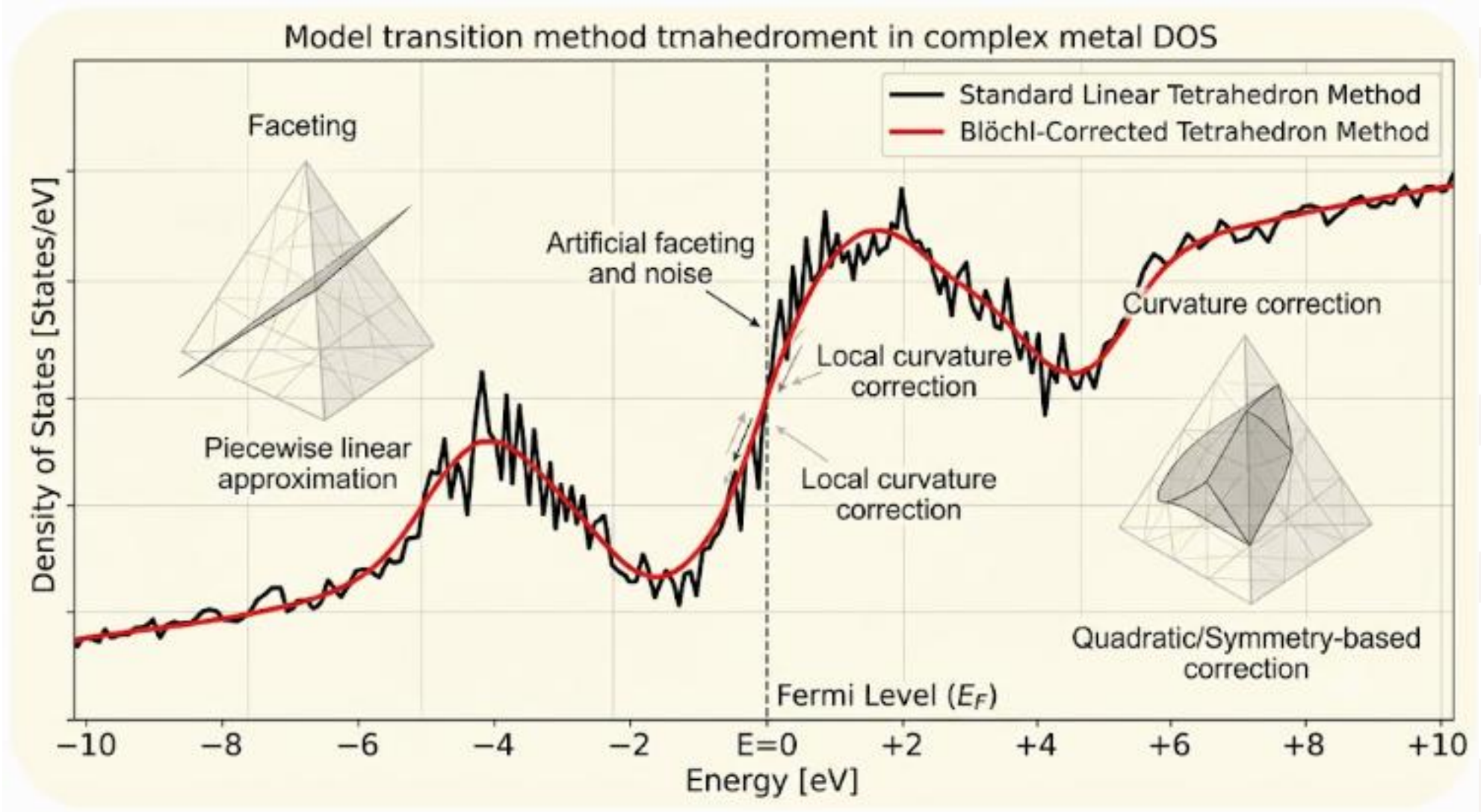
The Mechanism: Introduces local quadratic correction weights to account for the true curvature of the energy bands within each cell

This correction cancels out the systematic errors caused by linear interpolation without requiring a massive, unfeasible increase in the underlying grid density

To resolve these metallic integration errors, Peter Blöchl introduced a crucial optimization in 1994

The Blöchl-corrected tetrahedron method adds local correction weights that account for the true curvature of the energy bands within each cell. This modification effectively cancels out the systematic overestimation or underestimation caused by simple linear interpolation, allowing the algorithm to maintain high geometric accuracy even across sharp metallic Fermi surfaces

IV. Comparative Paradigms and Algorithmic Selection Guidelines



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Algorithmic Comparison Matrix

Evaluation Metric	Analytical Smearing Method	Tetrahedron Method (Blöchl)
Code Implementation	Extremely simple (Few lines of code)	Complex geometric sorting logic
Execution Speed	Fast; easily vectorizable loops	Slower due to cell-by-cell sorting
Artificial Parameters	Requires optimizing σ	Completely parameter-free
Van Hove Resolution	Blurred/Rounded edges	Perfectly sharp discontinuities
k Mesh Dependency	High; sensitive to grid spacing	Lower; converges faster with fewer points

This comparative matrix summarizes the engineering trade-offs between the two methods. Smearing methods are simple to implement and computationally efficient, but they require careful calibration of the parameter σ and tend to blur sharp physical features.

The tetrahedron method eliminates artificial parameters and captures Van Hove singularities with perfect geometric sharpness, but it involves more complex sorting logic and a higher computational overhead per grid cell. Choosing between them depends on whether the research task prioritizes execution speed or high edge resolution

IV. Comparative Paradigms and Algorithmic Selection Guidelines

Algorithmic Guidelines for Material Classification

Semiconductors & Insulators:
Recommendation: Use the Tetrahedron Method. It accurately captures band edges and preserves clean, state-free band gaps without artificial tail leakage

Complex Metals & Spintronics:
Recommendation: Use Methfessel-Paxton Smearing. It handles complex Fermi surface crossings efficiently and integrates smoothly over dense bands

High-Throughput Material Screening:
Recommendation: Use Standard Gaussian Smearing initially for rapid execution, then switch to the tetrahedron method for publication-grade figures

This guide establishes clear criteria for selecting the optimal numerical method based on the material system under study. For insulators and semiconductors, the tetrahedron method is highly recommended because it keeps the band gaps completely free of artificial states. For complex metals and transport calculations, high-order smearing schemes like Methfessel-Paxton are preferred due to their stability and speed when integrating over complex Fermi surfaces. High-throughput screening projects often combine both, using fast smearing for initial runs and the tetrahedron method for final, high-resolution plots



That's all for this session

Thank you for your attention and great participation